

DSC602 Information Visualization - July 2024 Exam Solutions

University of Buea | Faculty of Science | 70 Total Marks

PART 1: READING AND COMPREHENSION QUESTIONS (20 marks)

Question 1: What are the three visualization use cases? (6 marks)

The three primary visualization use cases are:

1. Explore (Exploratory Visualization) - 2 marks

Definition: The process of discovering patterns, relationships, outliers, and insights in data without predefined hypotheses.

Characteristics:

- Used when analysts don't know what to look for
- Driven by curiosity and investigation
- Interactive with multiple views
- Focus on data discovery

Example: Analyzing Titanic data to discover which passenger groups had higher survival rates, identifying age groups with suspicious patterns, finding correlations between variables.

Business Value: Generates new hypotheses and identifies unexpected trends that can guide further investigation.

2. Confirm (Confirmatory Visualization) - 2 marks

Definition: The process of validating or testing predetermined hypotheses using visualization as evidence.

Characteristics:

- Used when specific questions need answering
- Hypothesis-driven approach
- Focused visualization design
- Statistical validation support

Example: Testing the hypothesis "women and children had higher survival rates on the Titanic" by comparing survival rates across gender and age groups visually.

Business Value: Provides evidence-based confirmation or rejection of business hypotheses, supporting decision-making.

3. Present (Presentational Visualization) - 2 marks

Definition: The process of communicating findings and insights to stakeholders through clear, persuasive visualizations designed for comprehension.

Characteristics:

- Audience-focused design
- Clear messaging and storytelling
- Simplified for non-expert consumption
- High visual impact

Example: Creating a dashboard showing survival statistics by passenger class for executive presentation, highlighting key findings with annotations.

Business Value: Effectively communicates analytical findings to decision-makers, enabling informed business actions.

Question 2: Name a visualization technique for exploration/confirmation/presentation and describe it. (4 marks)

Visualization Technique: Scatter Plot (1 mark for naming)

For Exploration (Interactive Scatter Plot with Filtering):

Description:

A 2D plot showing individual data points positioned by two continuous variables (e.g., Age vs. Fare). Encoded with color for categorical variable (e.g., Survival status).

Why for Exploration:

- Reveals outliers immediately (passengers with very high fares, unusual ages)
- Shows clusters (first-class passengers group differently than third-class)
- Interactions (hover for details, brush to filter) enable discovery
- Multiple patterns visible simultaneously

Example Application: Explore Titanic data by plotting Age (x-axis) vs Fare (y-axis), colored by Survival status. This quickly reveals that higher fares correlate with better survival, and certain age groups behaved differently.

For Confirmation (Scatter Plot with Regression Line):

Description:

The same scatter plot with added trend line, confidence intervals, and statistical correlation coefficient.

Why for Confirmation:

- Visually validates the strength of relationships
- Regression line confirms expected trends
- Confidence interval shows uncertainty
- Statistical values provide evidence

Example Application: Confirm that "higher fare correlates with survival" by overlaying a trend line on the scatter plot, showing clear separation and R^2 value.

For Presentation (Annotated Scatter Plot):

Description:

Clean scatter plot with selected annotations, legend, title, axis labels, and highlights of key insights.

Why for Presentation:

- Focuses viewer attention on key findings
- Annotations explain significance
- Clean design maximizes comprehension
- Tells a clear story to non-technical audience

Example Application: Present finding "Passengers who paid higher fares had better survival chances" with annotations highlighting the high-fare survival cluster and statistics.

Question 3: What are the main visualization changes in visual data science? (8 marks)

Visual data science has undergone several major transformations:

1. From Static to Interactive (2 marks)

Historical: Fixed, printed visualizations with single perspective

- Limited to what designer anticipated
- No user interaction possible
- Expensive to create alternative views

Modern: Dynamic, interactive visualizations with multiple perspectives

- Users can filter, zoom, brush, hover
- Enables real-time exploration
- Responsive to user questions
- Examples: Dashboards, web-based analytics

Impact: Democratizes data exploration; non-technical users can discover insights independently.

2. From Aggregated to Individual (2 marks)

Historical: Summary statistics and aggregated plots (bar charts showing totals)

- Loss of detail in individual records
- Potential hiding of important patterns
- Anscombe's Quartet problem: different datasets with identical summaries

Modern: Individual data points visible with aggregation available

- Scatterplots showing every passenger's age/fare
- Linked views combining detail with summary

- Both individual anomalies AND patterns visible

Impact: More honest representations; individual outliers and edge cases now visible.

3. From Single View to Multiple Coordinated Views (2 marks)

Historical: Single visualization analyzed in isolation

- Difficult to relate multiple aspects of data
- Required mental integration of separate plots

Modern: Multiple linked visualizations (small multiples, brushing and linking)

- Change selection in one view highlights in others
- Age view linked to survival view linked to class view
- Holistic understanding through coordinated updates

Impact: Complex relationships become apparent through coordinate interaction.

4. From Designer-Driven to Data-Driven (2 marks)

Historical: Designer determines what to show and how

- Requires designer expertise
- Manual updates when data changes
- Fixed narratives

Modern: Automated visualization generation, Grammar of Graphics

- Specify what data to show, system determines best encoding
- Updates automatically as data changes
- Exploratory possibilities easily generated

Impact: Scales visualization creation; enables rapid iteration and experimentation.

Question 4: What is the difference between data visualization and information visualization? (2 marks)

Data Visualization (1 mark)

Definition: Visual representation of data using generic graphical techniques.

Characteristics:

- Focuses on displaying raw or processed data
- Generic techniques applicable to any dataset
- Examples: bar charts, line charts, scatter plots, pie charts
- Goal: Make data visible and interpretable
- Broader term; encompasses all visual representation

Domain: General-purpose; widely used in business, science, education

Example: Line chart showing sales over time, bar chart comparing regions, histogram showing age distribution.

Information Visualization (InfoVis) (1 mark)

Definition: Visual representation designed specifically for abstract, non-spatial data to reveal complex relationships and enable insight discovery.

Characteristics:

- Focuses on abstract information (not naturally spatial)
- Domain-specific and task-specific techniques
- Emphasizes interaction and exploration
- Goal: Support analytical reasoning and insight generation
- Narrower term; subset of visualization

Domain: Used for complex data analysis, business intelligence, scientific discovery

Example: Network visualization of social connections, treemap showing hierarchical data structure, parallel coordinates for multivariate analysis.

Key Differences:

Aspect	Data Visualization	Information Visualization
Data Type	Any data	Abstract, non-spatial data
Technique Specificity	Generic techniques	Domain/task-specific
Focus	Display and interpretation	Analytical reasoning
Interaction	Often static	Highly interactive
Scope	Broader category	More specialized subset
Example	Basic bar chart	Interactive dashboard

Relationship: Data Visualization is the broader field; Information Visualization is a specialized subset focused on abstract data and analytical discovery.

PART 2: PRACTICAL QUESTIONS - TITANIC DATASET (50 marks)

Question 1: What is the role of each step? (10 marks)

The data visualization process follows three key steps:

Step 1: Data Gathering (2 marks)

Role: Collect and consolidate data from various sources into a unified dataset.

Detailed Activities:

- **Data Collection:** Acquire raw data (CSV files, databases, APIs, sensors)

- Titanic example: Collect passenger records, survival information, ticket details
- **Data Extraction:** Pull relevant information from sources
 - Extract passenger lists, cabin assignments, fare information
- **Data Integration:** Combine data from multiple sources
 - Merge passenger details with embarkation records
- **Data Validation:** Ensure data integrity and completeness
 - Check for duplicate records, verify data ranges
- **Documentation:** Record data provenance and definitions
 - Note what each column represents, data collection methodology

Outcome: Unified, documented dataset ready for analysis (e.g., Titanic CSV with all passenger records).

Why Important: Poor data gathering leads to invalid conclusions; garbage in = garbage out.

Step 2: Data Analysis (5 marks)

Role: Explore, understand, and extract meaningful patterns from data.

Detailed Activities:

- **Data Cleaning:** Handle missing values, duplicates, outliers
 - Remove rows with missing cabin information
 - Identify and handle unusual values (negative ages, unrealistic fares)
- **Exploratory Data Analysis (EDA):** Understand data characteristics
 - Calculate summary statistics (mean, median, std dev)
 - Identify data distributions (age distribution skewed?)
 - Find relationships (do males or females survive more?)
- **Data Transformation:** Prepare data for visualization
 - Create derived variables (age groups from individual ages)
 - Aggregate data (sum deaths by embarkation port)
 - Normalize scales if necessary
- **Statistical Analysis:** Quantify relationships
 - Correlation between fare and survival
 - Chi-square test for independence between gender and survival
 - Calculate proportions (% women who survived)
- **Pattern Discovery:** Identify trends and anomalies
 - Which age groups had highest death rates?
 - Were first-class passengers more likely to survive?
 - Do certain cabins correlate with survival?

Outcome: Understood data characteristics, identified key patterns, prepared datasets for visualization.

Why Important: Analysis determines what to visualize and how; guides visualization design.

Step 3: Data Visualization (3 marks)

Role: Present data and analytical findings in visual form to enable human understanding and insight generation.

Detailed Activities:

- **Visualization Design:** Select appropriate techniques
 - Bar charts for comparisons (survival by gender)
 - Scatter plots for relationships (age vs fare)
 - Grouped visualizations for categorical comparisons
- **Visual Encoding:** Map data to visual properties
 - Position: Age on x-axis, death count on y-axis
 - Color: Gender as color (red=male, blue=female)
 - Shape: Different markers for different classes
- **Interactive Elements:** Enable exploration
 - Hover tooltips showing individual records
 - Filters for age ranges, passenger class
 - Linked views connecting different perspectives
- **Communication:** Tell the story
 - Add titles explaining what visualization shows
 - Annotate key findings
 - Provide legends and axis labels
- **Presentation:** Deliver to audience
 - Create dashboards for stakeholders
 - Generate reports with visualizations
 - Present findings clearly

Outcome: Visual representations of data that communicate insights and enable decision-making.

Why Important: Humans process visual information 60,000x faster than text; visualization makes complex patterns instantly apparent.

Relationship Between Steps (Cascade Effect):

1. **Data Gathering** → provides raw material for analysis
2. **Data Analysis** → identifies what patterns are important to show
3. **Data Visualization** → communicates those patterns effectively
4. **Feedback Loop:** Visualization may reveal need for further analysis or additional data

Question 2: Load and create a data frame (10 marks)

See code files for implementation (provided in DSC602_JULY_2024_TITANIC_SOLUTION.ipynb and .py files)

Key Requirements:

- Load Titanic dataset using pandas
- Display first 5 rows (head)
- Show total number of records
- Display shape and info about dataset

Theoretical Explanation (2 marks):

A DataFrame is the fundamental data structure for data analysis. It's a 2D table with rows (observations) and columns (variables). For the Titanic dataset:

- Rows: Individual passengers
 - Columns: Age, Sex, Survived, Fare, Embarked, Pclass, etc.
 - Benefits: Efficient computation, easy filtering, standard format for analysis
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Question 3: Remove rows with empty values (2 marks)

Role: Data cleaning step.

Why Important:

- Missing values can bias analysis (exclude entire passenger class if systematically missing)
- Most algorithms cannot handle NaN values
- Visualization becomes unclear with missing data

Process (See code):

1. Identify rows with any missing values
 2. Remove them
 3. Report number of records before and after
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Question 4: Compute cabin-related death statistics (6 marks)

Question Breakdown:

- **Part A (3 marks):** Count passengers NOT in cabin who DIED
- **Part B (3 marks):** Count passengers IN cabin who SURVIVED

Theoretical Insight:

This question tests understanding of:

- Boolean filtering (Where cabin is empty AND survived == 0)
 - Data aggregation (counting matching records)
 - Relationship between physical location and survival
 - Why having cabin information matters (proxies for location on ship)
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Question 5: Count unique cabins (3 marks)

Concept: Data exploration to understand data uniqueness.

Why Important:

- Shows how many distinct locations on ship
 - Indicates data sparsity (some cabins may appear once, others multiple times)
 - Useful for categorical analysis
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Question 6: Group by age and sex for death counts (5 marks)

Process:

1. Filter for deceased passengers (survived == 0)
2. Group by age AND sex
3. Count occurrences in each group
4. Display results

Theoretical Value:

- Tests understanding of multi-level grouping
 - Reveals demographic patterns (do certain age-gender combinations die more?)
 - Prepares data for visualization
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Question 7: Plot deaths by age and sex (5 marks)

Visualization Design:

- X-axis: Age
- Y-axis: Death count
- Color: Gender differentiation
- Type: Bar plot or scatter plot

Theoretical Concepts:

- Visual encoding: Use color channel for second dimension
 - Mark selection: Bars or points to show magnitudes
 - Effectiveness: Color is pre-attentive; viewers notice gender differences instantly
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Question 8: Ten ages with highest death count, differentiated by gender (5 marks)

Process:

1. Group deaths by age
2. Sort by count descending
3. Select top 10 ages
4. Create bar plot with gender differentiation

Visualization Technique:

Grouped or stacked bar chart where each bar represents an age, subdivided by gender using color.

Why Effective:

Shows both "which ages died most" AND "gender breakdown within each age" simultaneously.

Question 9: Deaths by embarkation port with gender breakdown (8 marks)

Concept: Three-way analysis (embarked port × gender × death count).

Process:

- 1. Group by embarkation port (S, C, Q)
- 2. Cross-tabulate with gender
- 3. Filter for deaths
- 4. Display results

Visualization Challenge:

Representing 3 dimensions simultaneously:

- Embarkation port (3 categories)
- Gender (2 categories)
- Death count (continuous)

Solution Approaches:

- Grouped bar chart: Ports on x-axis, gender-colored bars
- Faceted plots: Separate plot for each port
- Heatmap: Port × Gender with death count as color intensity

SUMMARY OF MARKS

Component	Marks	Status
Reading & Comprehension	20	Complete
Q1: Three use cases	6	✓
Q2: Visualization technique	4	✓
Q3: Main changes in visual data science	8	✓
Q4: Data vs Information Visualization	2	✓
Practical Questions	50	See code files
Q1: Role of steps (2+5+3)	10	See code
Q2: Load & DataFrame (2+2+6)	10	See code
Q3: Remove empty rows	2	See code
Q4: Cabin-related stats (3+3)	6	See code
Q5: Unique cabins	3	See code

Component	Marks	Status
Q6: Group by age & sex	5	See code
Q7: Plot deaths by age & sex	5	See code
Q8: Top 10 ages bar plot	5	See code
Q9: Deaths by embarkation & gender	8	See code
TOTAL	70	Complete

Exam Details:

- Date: Wednesday 10/07/2024
- Time: 12:00 - 15:00 (3 hours)
- Venue: CSC Lab
- Instructor: Dr. Nyamsi
- Course: DSC602 - Information Visualization
- Credit: 6

Code Implementation Notes:

- See [DSC602_JULY_2024_TITANIC_SOLUTION.ipynb](#) for Jupyter notebook with full code and visualizations
- See [DSC602_JULY_2024_TITANIC_SOLUTION.py](#) for standalone Python script
- Both files include detailed comments explaining each step and visualization choices
- Code is fully executable and produces all required outputs